

ROEBOURNE, Australia

WMO: 943090

Lat: 20.777S

Lon: 117.146E

Elev: 13

StdP: 101.17

Time Zone: 8.00 (E08)

Period: 94-05

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.9	14.1	-4.7	2.5	22.6	-2.7	3.0	22.3	11.8	18.1	9.2	19.1	0.6	270	0.333

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.3	42.8	23.8	41.4	23.5	40.1	23.6	28.4	35.4	27.7	34.4	27.2	33.4	2.3	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.8	22.5	31.1	26.2	21.7	30.4	25.8	21.1	30.0	91.9	35.4	88.6	34.6	86.0	33.4	36.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.8	7.1	5.4	9.8	45.5	0.7	1.4	9.3	46.5	8.9	47.3	8.6	48.1	8.1	49.1		
(5)			5.9	32.2	0.7	3.0	5.3	34.3	4.9	36.0	4.5	37.7	4.0	39.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.5	32.5	31.8	31.7	29.1	25.1	21.8	21.1	22.5	24.9	28.4	30.2	31.6		
	DBStd	4.72	2.43	2.55	2.29	1.99	2.23	2.39	1.81	1.91	2.45	3.00	2.61	2.52		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	3	0	0	0	0	0	1	2	0	0	0	0	0		
	CDD10.0	6401	697	611	672	572	467	353	345	389	448	571	606	671		
	CDD18.3	3362	438	377	414	322	208	104	88	131	198	313	356	413		
	CDH23.3	43125	6469	5459	6047	4037	1860	668	567	1163	2140	4022	4793	5899		
	CDH26.7	24542	4069	3286	3688	2174	747	162	113	389	1030	2351	2894	3637		
(14) Wind	WSAvg	1.8	1.7	1.8	1.6	1.4	1.7	1.9	1.8	1.8	1.8	1.9	2.0	2.0		
(15) Precipitation	PrecAvg	296	58	78	50	16	26	35	14	6	1	0	1	9		
	PrecMax	656	327	301	323	114	147	309	85	27	8	5	13	100		
	PrecMin	52	0	0	0	0	0	0	0	0	0	0	0	0		
	PrecStd	154	86	78	72	28	36	62	22	9	2	1	2	21		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.2	43.8	42.2	39.2	35.9	32.5	31.0	33.6	37.2	42.2	43.2	44.2		
		MCWB	24.2	24.6	23.6	22.0	20.0	20.4	17.5	18.6	19.8	21.5	22.9	23.2		
	2%	DB	42.5	41.2	40.5	37.5	33.8	30.3	29.3	31.7	35.7	40.0	41.4	42.2		
		MCWB	24.5	24.9	23.3	22.0	20.0	18.8	16.9	18.0	19.5	21.2	22.2	23.6		
	5%	DB	40.6	39.3	39.3	36.3	32.2	28.6	27.9	30.3	34.0	38.1	39.4	40.1		
		MCWB	24.7	24.9	23.5	21.4	19.9	18.1	16.7	17.1	19.1	20.6	22.1	24.0		
	10%	DB	38.7	37.4	37.8	35.0	30.7	27.2	26.6	29.0	32.3	36.1	37.7	38.1		
		MCWB	25.1	25.5	23.5	21.4	19.6	17.5	16.5	16.7	18.8	20.3	21.7	24.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.9	29.2	28.6	27.1	25.7	24.2	22.9	23.1	24.1	26.7	27.4	29.4		
		MCDB	36.8	35.7	35.5	35.6	29.6	26.8	25.0	26.3	33.4	33.0	36.8	36.1		
	2%	WB	28.1	28.3	27.6	26.0	24.2	22.4	21.1	20.8	21.7	24.1	25.2	27.7		
		MCDB	35.8	35.0	33.0	31.1	28.4	25.5	25.1	26.8	29.9	32.0	34.2	34.7		
	5%	WB	27.4	27.6	27.0	25.2	23.1	21.2	19.5	19.5	20.7	23.1	24.4	26.9		
		MCDB	34.6	33.9	32.0	30.8	27.7	25.4	24.5	26.6	29.6	31.6	33.5	33.5		
	10%	WB	26.8	27.1	26.4	24.6	22.2	20.1	18.2	18.5	19.9	22.2	23.6	26.2		
		MCDB	33.6	32.8	31.4	30.3	27.2	24.9	23.7	26.0	28.6	31.2	32.5	32.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.3	10.0	11.1	11.8	11.1	10.4	11.6	13.3	14.3	14.4	13.3	11.6		
		MCDBR	14.6	13.4	13.6	13.7	12.8	11.9	13.4	15.1	16.0	16.6	15.8	14.4		
	5% WB	MCWBR	4.1	4.1	4.4	4.5	4.8	4.9	5.4	5.8	5.7	5.4	5.4	4.8		
		MCWBR	11.5	10.3	10.6	9.9	10.2	8.8	10.7	12.6	14.7	14.3	12.9	10.4		
(40) Clear-Sky Solar Irradiance	taub		0.402	0.393	0.380	0.353	0.333	0.322	0.316	0.315	0.335	0.356	0.361	0.378		
	taud		2.494	2.532	2.556	2.584	2.572	2.590	2.583	2.541	2.487	2.457	2.483	2.511		
	Ebn at Noon		944	939	928	914	895	888	905	939	957	964	976	967		
	Edh at Noon		116	110	104	94	87	83	85	96	109	117	117	115		
(44) All-Sky Solar Radiation	RadAvg		7.44	7.11	6.36	5.56	4.66	4.22	4.58	5.66	6.68	7.55	8.04	8.01		
	RadStd		0.45	0.49	0.46	0.29	0.35	0.30	0.29	0.16	0.25	0.18	0.21	0.34		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)		+0.46	N/A	N/A	N/A	+0.48	+0.56	N/A	N/A	N/A	+158

Nomenclature: See separate page